Forces & Equilibrium

Section 1: Resolving Forces in Two Dimensions

1 A force of magnitude 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.

a. Horizontal: 43.3 N, Vertical: 25.0 N

b. Horizontal: 25.0 N, Vertical: 43.3 N

c. Horizontal: 35.4 N, Vertical: 35.4 N

d. Horizontal: 50.0 N, Vertical: 0.0 N

2 Two forces $F_1 = [8,\ -3]$ N and $F_2 = [-5,\ 7]$ N act on a particle. What is the magnitude of the resultant force?

a. 5 N

b. 7 N

c. $\sqrt{41}$ N

d. 12 N

3 A particle is held in equilibrium by three coplanar forces: P acting East, Q acting North, and a force of 13 N acting at an angle θ South of West where $\tan\theta = 5/12$. Find the magnitude of P.

a. 12 N

b. 5 N

c. 13 N

d. 8 N

Section 2: Friction & Limiting Equilibrium

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4 A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10\ \text{m/s}^2$, what is the maximum frictional force that can act on the block?

a. 21 N

b. 60 N

c. 18 N

d. 24 N

5 A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\max} = 21$ N). What is the actual frictional force acting on the block?

a. 15 N

b. 21 N

c. 0 N

d. 6 N

6 A particle of mass 4 kg is placed on a rough plane inclined at 30° to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction μ.

a. $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$

b. $\mu = \sin(30^\circ) = 0.5$

c. $\mu = \cos(30^\circ) \approx 0.866$

d. $\mu = 0.25$

Section 3: Inclined Planes & Equilibrium

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7 A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at 30° to the horizontal by a force P acting parallel to the slope. Taking $g = 10\ \text{m/s}^2$, find P.

a. 50 N

b. 86.6 N

c. 100 N

d. 25 N

8 For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?

a. 86.6 N ($50\sqrt{3}$ N)

b. 50 N

c. 100 N

d. 75 N

9 A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10\ \text{m/s}^2$, find the angle of inclination α of the plane.

a. $\alpha = \arccos(0.8) \approx 36.9^\circ$

b. $\alpha = 30^\circ$

c. $\alpha = 45^\circ$

d. $\alpha = 60^\circ$

Section 4: Lami's Theorem & Connected Systems

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10 A lamp of weight $W = 60$ N is suspended by two light strings making angles of 30° and 60° with the horizontal ceiling. Find the tension in the steeper string.

a. $30\sqrt{3}$ N ≈ 52.0 N

b. 30 N

c. 60 N

d. 45 N

11 A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s^2)

a. 20 N

b. 50 N

c. 30 N

d. 10 N

12 A ring of mass m is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension T at an angle of 45° above the wire. Find the value of T (in terms of mg) when the ring is in limiting equilibrium.

a. $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$

b. $\dfrac{\sqrt{2}}{2}mg$

c. $0.4mg$

d. mg

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